

Application Deployment

(Deploy the given react application to a production ready state)

Application:

Clone the below mentioned repo and deploy the application.
(Run the application in port 80 [HTTP])

Repo URL: <https://github.com/sriram-R-krishnan/devops-build>

Docker:

- Dockerize the application by creating a Dockerfile
- Create a docker-compose file to use the above image

Bash Scripting:

- Write 2 scripts
 - **build.sh** for building docker images
 - **deploy.sh** for deploying the image to server

Version Control:

- Push the code to github to **dev branch** (use **dockerignore** & **gitignore** files)
- **Note:** Use only CLI for related git commands

Docker hub:

- Create 2 repos "**dev**" and "**prod**" to push images.
- "**Prod**" repo must be private and "**dev**" repo can be public

Jenkins:

- Install and configure jenkins build step as per needs to build, push & deploy the application
- Connect jenkins to the github repo with auto build trigger from both **dev** & **master** branch
- If code pushed to **dev branch**, docker image must build and pushed to **dev repo** in docker hub
- If **dev merged to master**, then docker image must be pushed to **prod repo** in docker hub

AWS:

- Launch **t2.micro instance** and deploy the create application.
- Configure SG as below:
 - Whoever has the ip address can access the application
 - Login to server can should be made only from your ip address

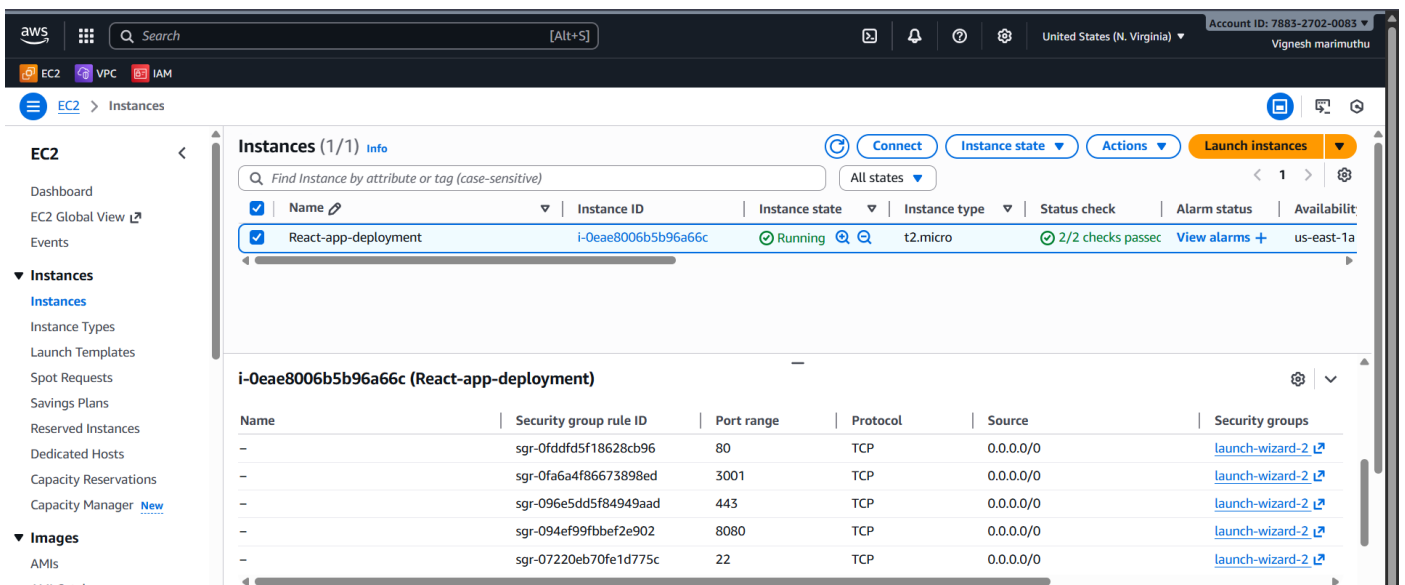
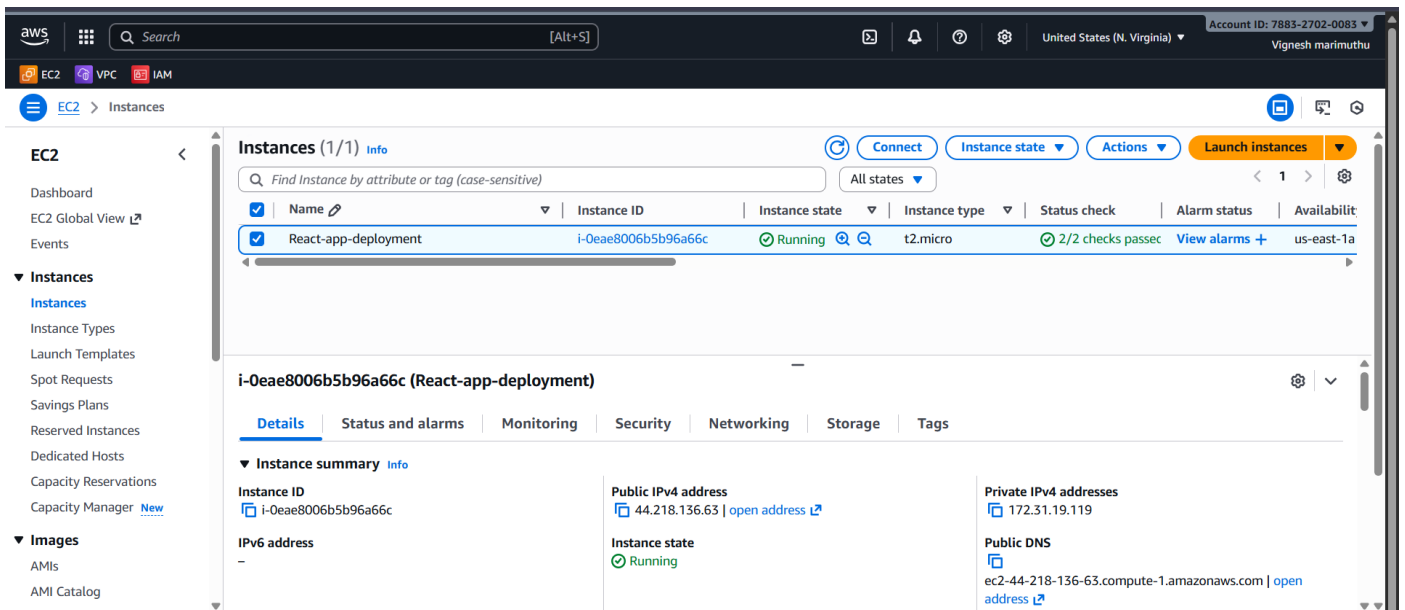
Monitoring:

- Setup a monitoring system to check the health status of the application. (Open-source)
- Sending notifications only if the application goes down is highly appreciable

Solution:

✓ 1. Launched an EC2 instance and connected to Ubuntu via SSH

- EC2 will act as the *Jenkins build server* and *control center* for CI/CD, Docker builds, etc.



- Security group allows inbound traffic on ports **80 (HTTP)**, **3001 (Uptime Kuma)**, **443 (HTTPS)**, **8080 (Jenkins)**, and **22 (SSH)** for web access, application access, and secure remote management.

```
Ubuntu-24.04
vignesh@vignesh:~$ ssh -i /home/vignesh/mykey.pem ubuntu@44.218.136.63
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1015-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Thu Nov  6 19:19:01 UTC 2025

System load:  0.14           Processes:            112
Usage of /:   25.9% of 6.71GB Users logged in:             0
Memory usage: 21%           IPv4 address for enX0: 172.31.19.119
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

Last login: Thu Nov  6 19:19:03 2025 from 106.219.182.112
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-172-31-19-119:~$
```

✓ 2. Prepare the React repository (locally)

```
ubuntu@ip-172-31-19-119:~$ git clone https://github.com/sriram-R-krishnan/devops-build.git
cd devops-build
Cloning into 'devops-build'...
remote: Enumerating objects: 21, done.
remote: Total 21 (delta 0), reused 0 (delta 0), pack-reused 21 (from 1)
Receiving objects: 100% (21/21), 720.09 KiB | 16.75 MiB/s, done.
ubuntu@ip-172-31-19-119:~/devops-build$ git remote remove origin
ubuntu@ip-172-31-19-119:~/devops-build$ git remote add origin https://github.com/vigneshmarimuthu/React-App-CI-CD.git
ubuntu@ip-172-31-19-119:~/devops-build$ git checkout -b dev
Switched to a new branch 'dev'
```

Created .dockerignore & .gitignore

- .gitignore avoids huge or secret files being committed.
- .dockerignore avoids sending unnecessary files to the Docker daemon during build.

```
Switched to a new branch 'dev'
ubuntu@ip-172-31-19-119:~/devops-build$ cat > .gitignore <<'EOF'
node_modules/
.env
dist/
.DS_Store
EOF

cat > .dockerignore <<'EOF'
node_modules
.git
npm-debug.log
Dockerfile*
EOF
ubuntu@ip-172-31-19-119:~/devops-build$
```

Created Dockerfile

- Used a lightweight web server image (Nginx) and copy only the built static files into it.

This makes our final image very small and secure.

```
ubuntu@ip-172-31-19-119:~/devops-build$ cat Dockerfile
# Use the official Nginx image to serve static files
FROM nginx:alpine

# Copy all static files from your project directory to Nginx HTML folder
COPY . /usr/share/nginx/html

# Expose port 80
EXPOSE 80

# Start Nginx
CMD ["nginx", "-g", "daemon off;"]
ubuntu@ip-172-31-19-119:~/devops-build$
```

Created docker-compose.yml

- `docker-compose up -d` provides convenient local testing

```
ubuntu@ip-172-31-19-119: ~/devops-build$ nano docker-compose.yml
ubuntu@ip-172-31-19-119: ~/devops-build$ cat docker-compose.yml
version: '3.8'
services:
  web:
    image: my-dockerhub-user/dev:latest
    # pipeline will update tags or use image pull
    ports:
      - "80:80"
    restart: always
ubuntu@ip-172-31-19-119: ~/devops-build$
```

✓ 3. Added build & deploy scripts

- These scripts will be used later by Jenkins.
- Make the scripts executable.

```
ubuntu@ip-172-31-19-119: ~/devops-build$ nano build.sh
ubuntu@ip-172-31-19-119: ~/devops-build$ cat build
cat: build: Is a directory
ubuntu@ip-172-31-19-119: ~/devops-build$ cat build.sh
#!/usr/bin/env bash
set -e
set -o pipefail
# Usage: ./build.sh <dockerhub_user> <repo> <tag>
DOCKER_USER=${1:-yourdockeruser}
REPO=${2:-dev}
TAG=${3:-latest}
IMAGE="${DOCKER_USER}/${REPO}:${TAG}"
echo "Building image $IMAGE"
# Build using local Dockerfile
docker build -t "$IMAGE" .
echo "Pushing image to Docker Hub"
docker push "$IMAGE"
```

```
ubuntu@ip-172-31-19-119: ~/devops-build$ nano deploy.sh
ubuntu@ip-172-31-19-119: ~/devops-build$ cat deploy.sh
#!/usr/bin/env bash
set -e
set -o pipefail
DOCKER_USER=${1:-yourdockeruser}
REPO=${2:-dev}
TAG=${3:-latest}
IMAGE="${DOCKER_USER}/${REPO}:${TAG}"

# If using docker-compose, pull and recreate
if [ -f docker-compose.yml ]; then
  echo "Pulling image $IMAGE"
  docker-compose pull || true
  docker-compose up -d --remove-orphans
else
  # fallback to docker run
  echo "Stopping old container (if any)"
  docker ps -q --filter "name=webapp" | xargs -r docker stop
  docker ps -a -q --filter "name=webapp" | xargs -r docker rm
  echo "Running new container"
  docker run -d --name webapp -p 80:80 --restart=always "$IMAGE"
fi

echo "Deploy complete"
ubuntu@ip-172-31-19-119: ~/devops-build$
```

✓ 4.Pushed to my GitHub repo

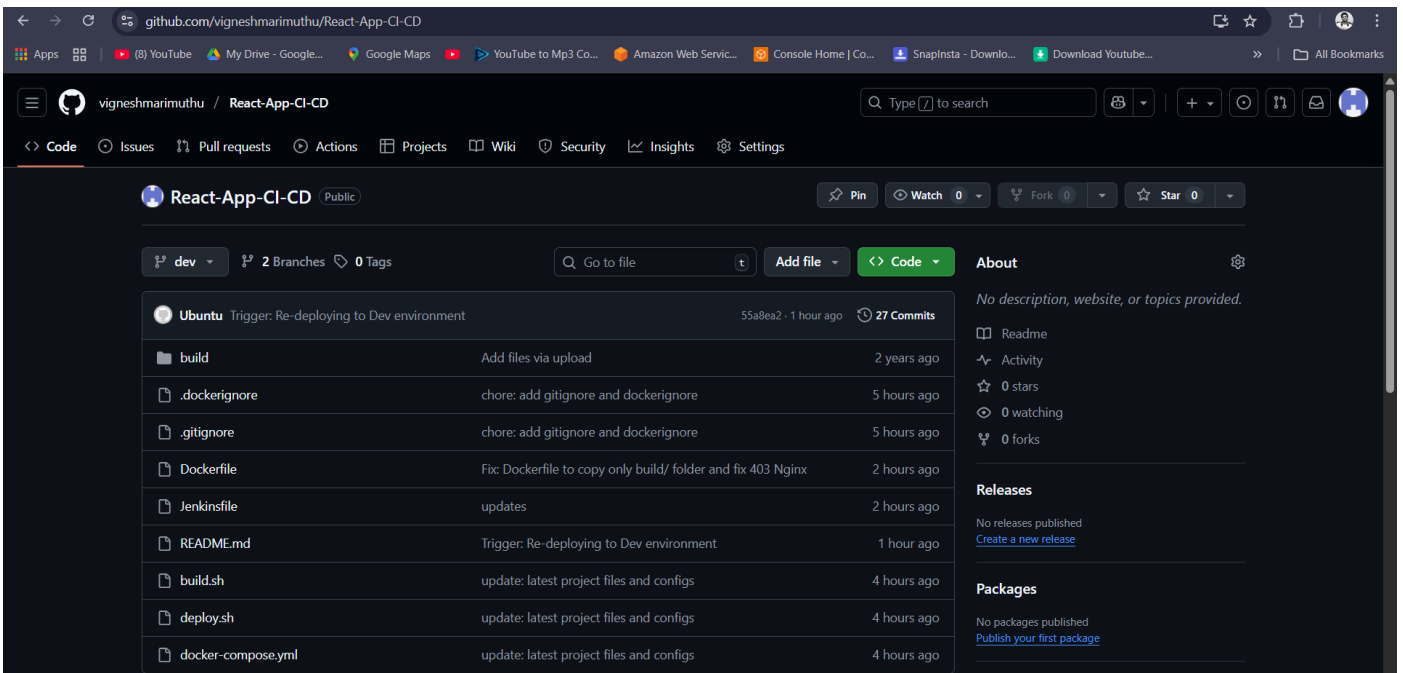
```
ubuntu@ip-172-31-19-119: ~/devops-build$ ls -a
. .dockerignore .git .gitignore Dockerfile build build.sh deploy.sh docker-compose.yml
ubuntu@ip-172-31-19-119: ~/devops-build$ git checkout dev
Already on 'dev'
Your branch is up to date with 'origin/dev'.
ubuntu@ip-172-31-19-119: ~/devops-build$ git add .
git commit -m "update: latest project files and configs"
git push origin dev
[dev b31a4a6] update: latest project files and configs
Committer: Ubuntu <ubuntu@ip-172-31-19-119.ec2.internal>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly. Run the
following command and follow the instructions in your editor to edit
your configuration file:

    git config --global --edit

After doing this, you may fix the identity used for this commit with:

    git commit --amend --reset-author

4 files changed, 55 insertions(+)
create mode 100644 Dockerfile
create mode 100755 build.sh
create mode 100755 deploy.sh
create mode 100644 docker-compose.yml
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Compressing objects: 100% (6/6), done.
Writing objects: 100% (6/6), 1.40 KiB | 1.40 MiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0
To github.com:vigneshmarimuthu/React-App-CI-CD.git
   cc73ecf..b31a4a6 dev -> dev
ubuntu@ip-172-31-19-119: ~/devops-build$
```



The screenshot shows the GitHub repository page for **vigneshmarimuthu / React-App-CI-CD**. The repository is public and has 0 stars, 0 forks, and 0 releases. The main content area displays a list of files and their commit history:

File	Commit Message	Time
build	Add files via upload	2 years ago
.dockerignore	chore: add gitignore and dockerignore	5 hours ago
.gitignore	chore: add gitignore and dockerignore	5 hours ago
Dockerfile	Fix: Dockerfile to copy only build/ folder and fix 403 Nginx	2 hours ago
Jenkinsfile	updates	2 hours ago
README.md	Trigger: Re-deploying to Dev environment	1 hour ago
build.sh	update: latest project files and configs	4 hours ago
deploy.sh	update: latest project files and configs	4 hours ago
docker-compose.yml	update: latest project files and configs	4 hours ago

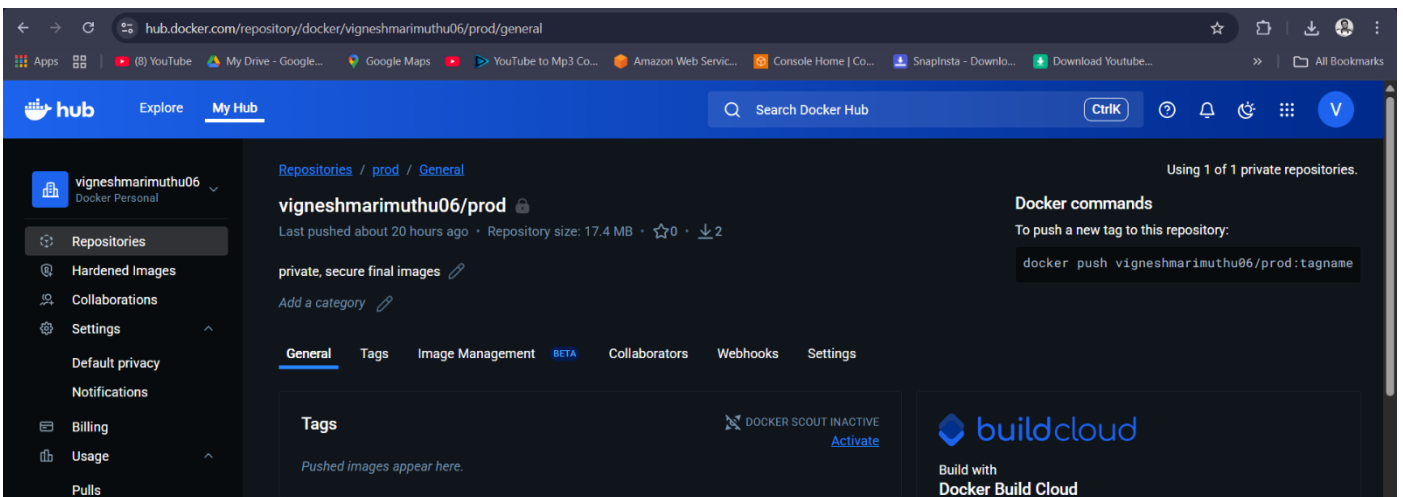
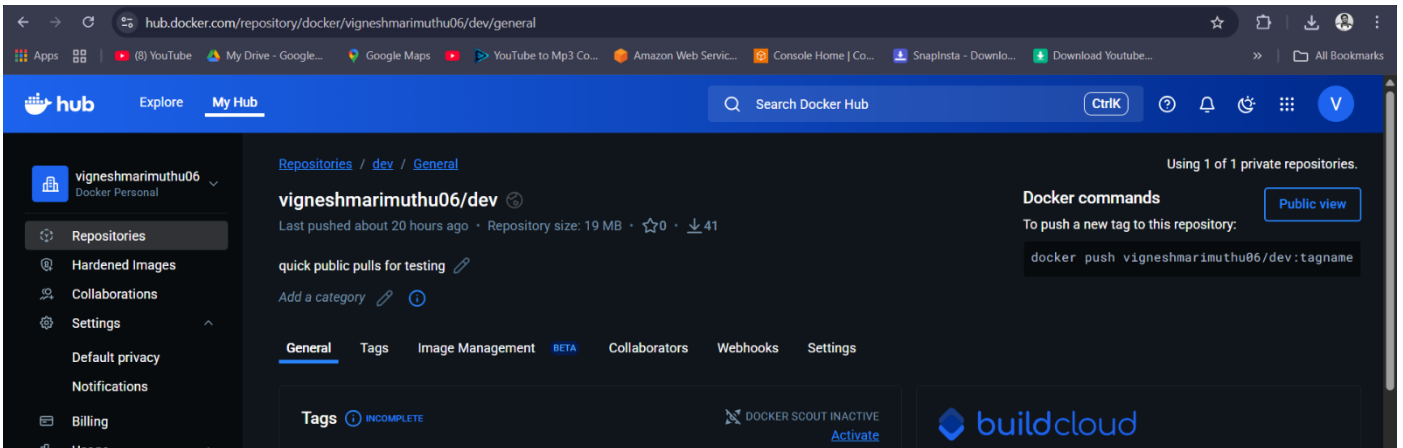
The sidebar on the right contains the following sections:

- About**: No description, website, or topics provided.
- Releases**: No releases published. [Create a new release](#)
- Packages**: No packages published. [Publish your first package](#)

- My GitHub repository: <https://github.com/vigneshmarimuthu/React-App-CI-CD>

✓ 5. Create Docker Hub repositories

- Dev repo: quick public pulls for testing. Prod repo: private, secure final images.



✓ 6. Set up the EC2 Ubuntu server: install Docker and Docker Compose

- Jenkins (running in a container) will use the host Docker socket to build images.

```
ubuntu@ip-172-31-19-119:~/devops-build$ # [ ] Update system
sudo apt update -y
sudo apt upgrade -y

# [ ] Install prerequisites
sudo apt install -y ca-certificates curl gnupg lsb-release

# [ ] Add Docker's official GPG key
sudo mkdir -p /etc/apt/keyrings
curl -fsSL https://download.docker.com/linux/ubuntu/gpg | sudo gpg --dearmor -o /etc/apt/keyrings/docker.gpg

# [ ] Add Docker repository
echo \
"deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.gpg] \
https://download.docker.com/linux/ubuntu \
$(lsb_release -cs) stable" | \
sudo tee /etc/apt/sources.list.d/docker.list > /dev/null

# [ ] Install Docker Engine, CLI, and Compose plugin
sudo apt update -y
sudo apt install -y docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin

# [ ] Start and enable Docker
sudo systemctl start docker
sudo systemctl enable docker

# [ ] Add your user (ubuntu) to the docker group
sudo usermod -aG docker $USER
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
```

```
ubuntu@ip-172-31-19-119: ~/devops-build$ sudo apt update
sudo apt install docker-compose -y
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:4 https://download.docker.com/linux/ubuntu noble InRelease
Ign:5 https://pkg.jenkins.io/debian-stable binary/ InRelease
Hit:6 https://pkg.jenkins.io/debian-stable binary/ Release
Get:7 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:9 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1305 kB]
Get:10 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [215 kB]
Get:11 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [2135 kB]
Get:12 http://security.ubuntu.com/ubuntu noble-security/restricted Translation-en [485 kB]
Fetched 4265 kB in 4s (1136 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
```

✓ 7. Installed and run Jenkins (in Docker) on EC2

- `-v /var/run/docker.sock:/var/run/docker.sock` lets Jenkins control the host Docker (build/push images) from inside container.
- This is standard and widely used, but it grants elevated access to Docker on the host — secure the server.

```
ubuntu@ip-172-31-19-119: $ #❏ Update and install dependencies
sudo apt update -y
sudo apt install -y fontconfig openjdk-17-jre curl gnupg

#❏ Add the Jenkins GPG key properly
sudo mkdir -p /usr/share/keyrings
curl -fsSL https://pkg.jenkins.io/debian-stable/jenkins.io-2023.key | sudo tee /usr/share/keyrings/jenkins-keyring.asc > /dev/null

#❏ Add the Jenkins repository
echo "deb [signed-by=/usr/share/keyrings/jenkins-keyring.asc] https://pkg.jenkins.io/debian-stable binary/" | \
  sudo tee /etc/apt/sources.list.d/jenkins.list > /dev/null

#❏ Update package list
sudo apt update -y

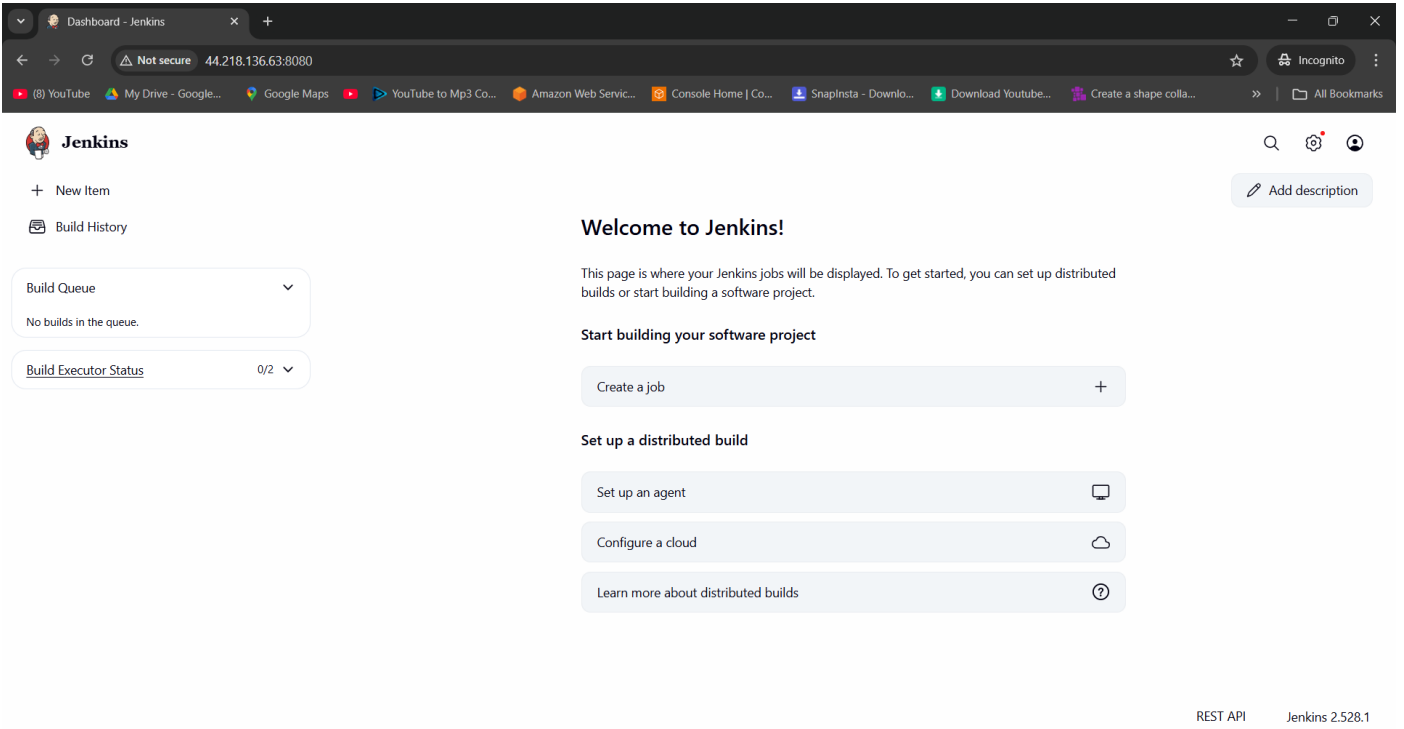
#❏ Install Jenkins
sudo apt install -y jenkins

#❏ Start and enable Jenkins service
sudo systemctl enable jenkins
sudo systemctl start jenkins
sudo systemctl status jenkins
```

```
No VM guests are running outdated hypervisor (qemu) binaries on this host.
Synchronizing state of jenkins.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable jenkins
• jenkins.service - Jenkins Continuous Integration Server
   Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset: enabled)
   Active: active (running) since Thu 2025-11-06 19:37:07 UTC; 6s ago
   Main PID: 10317 (java)
     Tasks: 45 (limit: 1121)
    Memory: 319.0M (peak: 354.8M)
       CPU: 16.009s
   CGroup: /system.slice/jenkins.service
           └─10317 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.war --webroot=/var/cache/jenkins/war --httpPort=8080

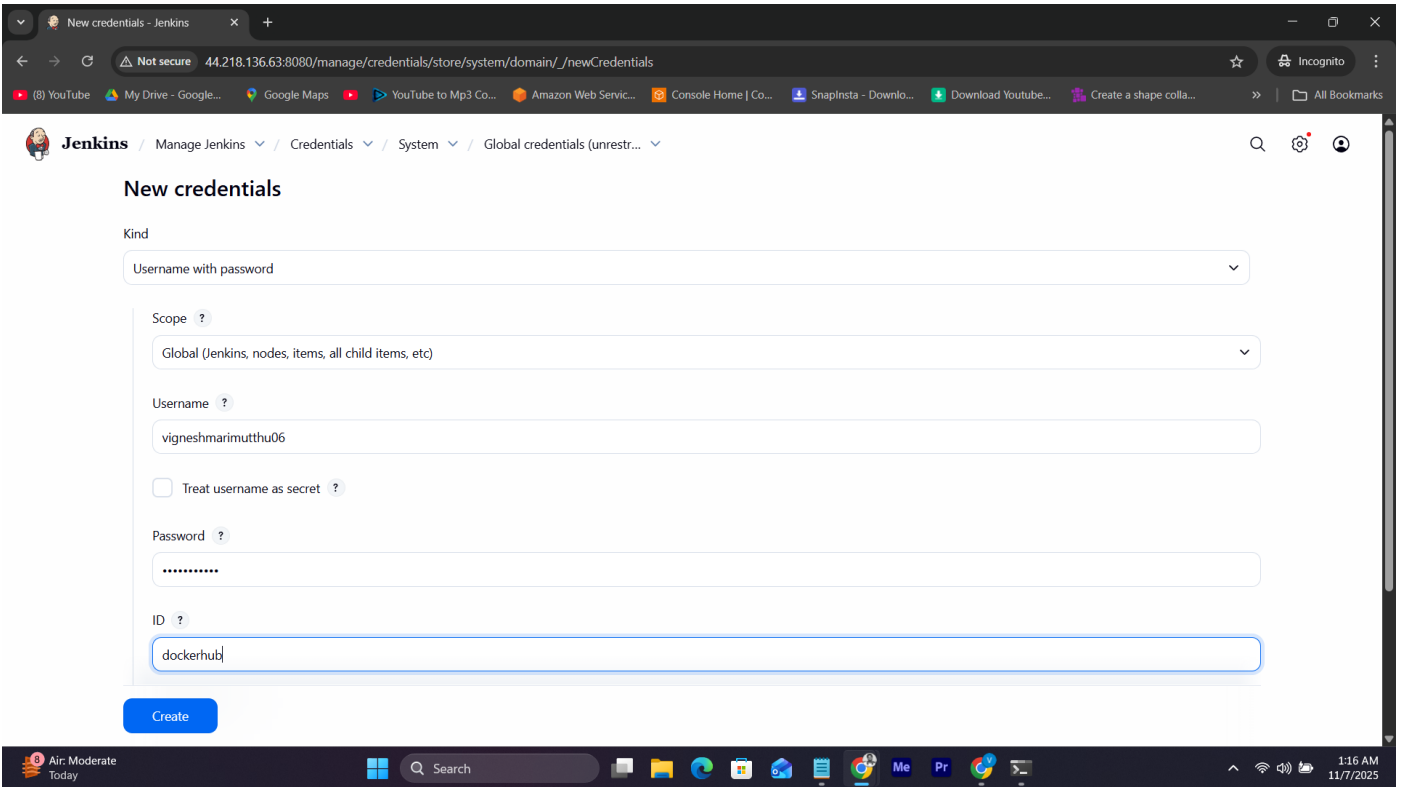
Nov 06 19:37:00 ip-172-31-19-119 jenkins[10317]: [LF]> This may also be found at: /var/lib/jenkins/secrets/initialAdminPassword
Nov 06 19:37:00 ip-172-31-19-119 jenkins[10317]: [LF]>
Nov 06 19:37:00 ip-172-31-19-119 jenkins[10317]: [LF]> *****
Nov 06 19:37:00 ip-172-31-19-119 jenkins[10317]: [LF]> *****
Nov 06 19:37:00 ip-172-31-19-119 jenkins[10317]: [LF]> *****
Nov 06 19:37:07 ip-172-31-19-119 jenkins[10317]: 2025-11-06 19:37:07.583+0000 [id=30] INFO jenkins.InitReactorRunner$1#onAttained: Completed >
Nov 06 19:37:07 ip-172-31-19-119 jenkins[10317]: 2025-11-06 19:37:07.623+0000 [id=23] INFO hudson.lifecycle.Lifecycle#onReady: Jenkins is ful
Nov 06 19:37:07 ip-172-31-19-119 systemd[1]: Started jenkins.service - Jenkins Continuous Integration Server.
Nov 06 19:37:07 ip-172-31-19-119 jenkins[10317]: 2025-11-06 19:37:07.931+0000 [id=47] INFO h.m.DownloadService$Downloadable#load: Obtained the
Nov 06 19:37:07 ip-172-31-19-119 jenkins[10317]: 2025-11-06 19:37:07.933+0000 [id=47] INFO hudson.util.Retrier#start: Performed the action ch
lines 1-20/20 (END)
```

✓ 8. Opened Jenkins UI



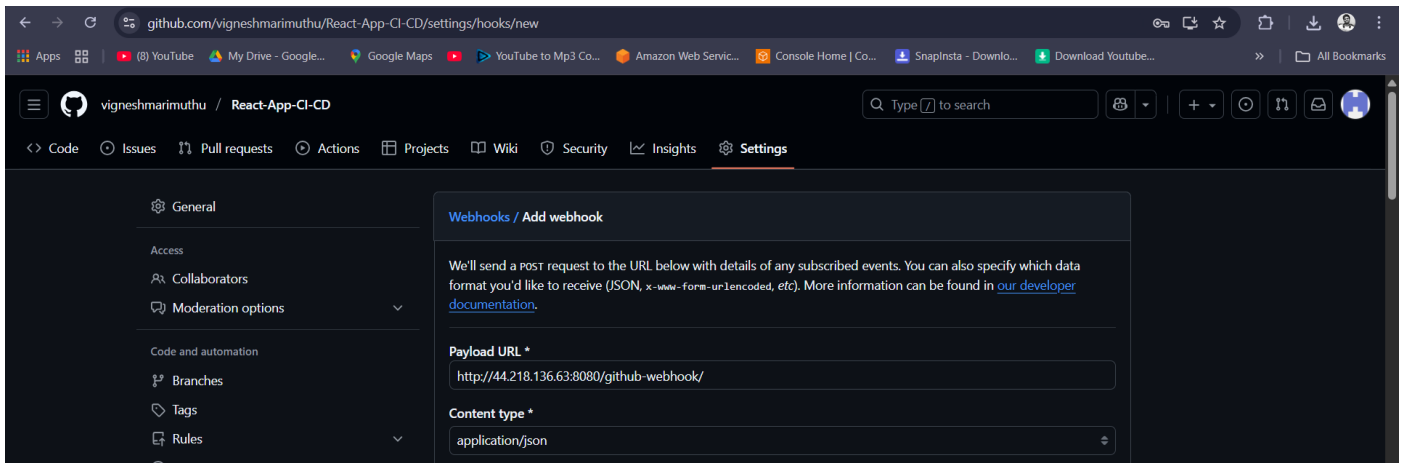
The screenshot shows the Jenkins Dashboard in a web browser. The browser's address bar displays "Not secure" and the URL "44.218.136.63:8080". The Jenkins logo is in the top left corner, and the page title is "Dashboard - Jenkins". On the left sidebar, there are links for "New Item", "Build History", "Build Queue", and "Build Executor Status". The "Build Queue" section shows "No builds in the queue." and the "Build Executor Status" shows "0/2". The main content area has a heading "Welcome to Jenkins!" followed by a paragraph: "This page is where your Jenkins jobs will be displayed. To get started, you can set up distributed builds or start building a software project." Below this is a section "Start building your software project" with a "Create a job" button. Further down is a section "Set up a distributed build" with three buttons: "Set up an agent", "Configure a cloud", and "Learn more about distributed builds". At the bottom right, there are links for "REST API" and "Jenkins 2.528.1".

Jenkins: added credentials (Docker Hub)



The screenshot shows the "New credentials" form in the Jenkins web interface. The browser's address bar displays "Not secure" and the URL "44.218.136.63:8080/manage/credentials/store/system/domain/_/newCredentials". The Jenkins logo is in the top left corner, and the page title is "New credentials". The form has a "Kind" dropdown menu set to "Username with password". Below this is a "Scope" dropdown menu set to "Global (Jenkins, nodes, items, all child items, etc)". The "Username" field contains the text "vigneshmarimutthu06". There is a checkbox labeled "Treat username as secret" which is unchecked. The "Password" field is masked with "*****". The "ID" field contains the text "dockerhub". At the bottom of the form is a blue "Create" button. The Windows taskbar is visible at the bottom of the screen, showing the time as 11:16 AM on 11/7/2025.

Configure GitHub webhook → Jenkins



✅ 9. Created Jenkinsfile

```
ubuntu@ip-172-31-19-119: ~/devops-build$ nano Jenkinsfile
ubuntu@ip-172-31-19-119: ~/devops-build$ cat Jenkinsfile
pipeline {
    agent any

    environment {
        DOCKER_USER = 'vigneshmarimuthu06'
        // Jenkins credential ID for DockerHub (Username + Password or PAT)
        CREDENTIAL_ID = 'dockerhub'
    }

    stages {
        stage('Checkout') {
            steps {
                checkout scm
            }
        }

        stage('Build and Push Image') {
            steps {
                script {
                    withCredentials([usernamePassword(credentialsId: CREDENTIAL_ID, usernameVariable: 'DOCKER_USER', passwordVariable: 'DOCKER_PASS')]) {
                        // Determine branch
                        def branchName = env.BRANCH_NAME ? sh(script: "git rev-parse --abbrev-ref HEAD", returnStdout: true).trim()

                        if (branchName == "dev") {
                            echo "🔍 Detected branch: dev – Building and pushing Dev image..."
                            sh '''
                                echo $DOCKER_PASS | docker login -u $DOCKER_USER --password-stdin
                                docker build -t $DOCKER_USER/dev:latest .
                                docker push $DOCKER_USER/dev:latest
                                docker logout
                            '''
                        } else if (branchName == "main" || branchName == "master") {
                            echo "🔍 Detected branch: main/master – Building and pushing Prod image..."
                            sh '''
```

```
stage('Build and Push Image') {
  steps {
    script {
      withCredentials([usernamePassword(credentialsId: CREDENTIAL_ID, usernameVariable: 'DOCKER_USER', passwordVariable: 'DOCKER_PASS')]) {
        // Determine branch
        def branchName = env.BRANCH_NAME ?: sh(script: "git rev-parse --abbrev-ref HEAD", returnStdout: true).trim()

        if (branchName == "dev") {
          echo "🔍 Detected branch: dev - Building and pushing Dev image..."
          sh '''
            echo $DOCKER_PASS | docker login -u $DOCKER_USER --password-stdin
            docker build -t $DOCKER_USER/dev:latest .
            docker push $DOCKER_USER/dev:latest
            docker logout
          '''
        } else if (branchName == "main" || branchName == "master") {
          echo "🔍 Detected branch: main/master - Building and pushing Prod image..."
          sh '''
            echo $DOCKER_PASS | docker login -u $DOCKER_USER --password-stdin
            docker build -t $DOCKER_USER/prod:latest .
            docker push $DOCKER_USER/prod:latest
            docker logout
          '''
        } else {
          echo "⚠️ Branch '${branchName}' not configured for Docker push. Skipping image push."
        }
      }
    }
  }
}

post {
  always {
    echo "✅ Pipeline finished for branch: ${env.BRANCH_NAME}"
  }
}
```

ubuntu@ip-172-31-19-119: ~/devops-build\$

✅ 10. Jenkins Pipeline

New Item - Jenkins

44.218.136.63:8080/view/all/newJob

(8) YouTube | My Drive - Google... | Google Maps | YouTube to Mp3 Co... | Amazon Web Servic... | Console Home | Co... | SnapInsta - Downlo... | Download Youtube... | Create a shape colla... | All Bookmarks

Jenkins / All / New Item

🔍 ⚙️ 👤

New Item

Enter an item name

react-app-pipeline

Select an item type

 **Freestyle project**
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.

 **Pipeline**
Orchestrates long-running activities that can span multiple build agents. Suitable for building pipelines (formerly known as workflows) and/or organizing complex activities that do not easily fit in free-style job type.

 **Multi-configuration project**
Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.

 **Folder**
Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the same name as long as they are in different folders.

OK

react-app-pipeline Config - Jen x +

Not secure 44.218.136.63:8080/job/react-app-pipeline/configure

Jenkins / react-app-pipeline / Configuration

Configure

- General
- Triggers
- Pipeline
- Advanced

Pipeline

Define your Pipeline using Groovy directly or pull it from source control.

Definition

Pipeline script from SCM

SCM ?

Git

Repositories ?

Repository URL ?

https://github.com/vigneshmarimuthu/React-App-CI-CD

Credentials ?

vigneshmarimuthu06/*****

+ Add

Advanced

Save Apply

react-app-pipeline Config - Jen x +

Not secure 44.218.136.63:8080/job/react-app-pipeline/configure

Jenkins / react-app-pipeline / Configuration

Configure

- General
- Triggers
- Pipeline
- Advanced

Pipeline

+ Add Repository

Branches to build ?

Branch Specifier (blank for 'any') ?

*/dev

+ Add Branch

Repository browser ?

(Auto)

Additional Behaviours

+ Add

Script Path ?

Jenkinsfile

Save Apply

```
ubuntu@ip-172-31-19-119:~/devops-build$ echo "# React-App-CI-CD"
This repository contains a CI/CD setup for a React application using Jenkins, Docker, and AWS EC2.
```

```
Trigger test: $(date)
" > README.md
ubuntu@ip-172-31-19-119:~/devops-build$ git add .
git commit -m "test: trigger pipeline build"
[dev 54b1164] test: trigger pipeline build
Committer: Ubuntu <ubuntu@ip-172-31-19-119.ec2.internal>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly. Run the
following command and follow the instructions in your editor to edit
your configuration file:
```

```
git config --global --edit
```

After doing this, you may fix the identity used for this commit with:

```
git commit --amend --reset-author
```

```
1 file changed, 5 insertions(+)
create mode 100644 README.md
ubuntu@ip-172-31-19-119:~/devops-build$ git push origin dev
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 430 bytes | 430.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:vigneshmarimuthu/React-App-CI-CD.git
51dc8d7..54b1164 dev -> dev
ubuntu@ip-172-31-19-119:~/devops-build$
```

The screenshot shows the Jenkins web interface for a pipeline named 'React-pipeline'. The 'Console Output' tab is selected, displaying the execution log of a build triggered by a GitHub push. The log shows the following steps:

- Started by GitHub push by vigneshmarimuthu
- Checking out git repository: <https://github.com/vigneshmarimuthu/React-App-CI-CD>
- Selected Git installation does not exist. Using Default
- The recommended git tool is: NONE
- using credential dockerhub
- git rev-parse --resolve-git-dir /var/lib/jenkins/workspace/React-pipeline@script/27b1c8f02105ee5cd58c9bf782d4cd3b5bad0a80c9450ca3bd03d87b59847939/.git # timeout=10
- Fetching changes from the remote Git repository
- git config remote.origin.url https://github.com/vigneshmarimuthu/React-App-CI-CD # timeout=10
- Fetching upstream changes from https://github.com/vigneshmarimuthu/React-App-CI-CD
- git --version # timeout=10
- git --version # 'git version 2.43.0'
- using GIT_ASKPASS to set credentials
- git fetch --tags --force --progress -- https://github.com/vigneshmarimuthu/React-App-CI-CD +refs/heads/*:refs/remotes/origin/* # timeout=10
- Seen branch in repository origin/dev
- Seen branch in repository origin/main
- Seen 2 remote branches
- git show-ref --tags -d # timeout=10
- Checking out Revision cea3d0999bcbef70dd14c547d25675f79558ea8 (origin/main)
- git config core.sparsecheckout # timeout=10
- git checkout -f cea3d0999bcbef70dd14c547d25675f79558ea8 # timeout=10
- Commit message: "Fix: Dockerfile to copy only build/ folder and fix 403 Nginx"
- git rev-list --no-walk b1cfa3bddb86a78e1e3ad92755b084379a4be27e # timeout=10

The screenshot shows the Docker Hub repository page for 'vigneshmarimuthu06/dev'. The repository is a Docker image with a size of 41.3 MB and 41 stars. The 'General' tab is selected, showing the repository details and tags.

Tags

Tag	OS	Type	Pulled	Pushed
latest	linux	Image	less than 1 day	less than a minute

[See all](#)

buildcloud

Build with Docker Build Cloud

Accelerate image build times with access to cloud-based builders and shared cache.

Docker Build Cloud executes builds on optimally-dimensioned cloud infrastructure with dedicated per-organization isolation.

```

ubuntu@ip-172-31-19-119: ~/devops-build$ # Switch to main branch
git checkout main

# Pull the latest version of main from GitHub (just to be safe)
git pull origin main

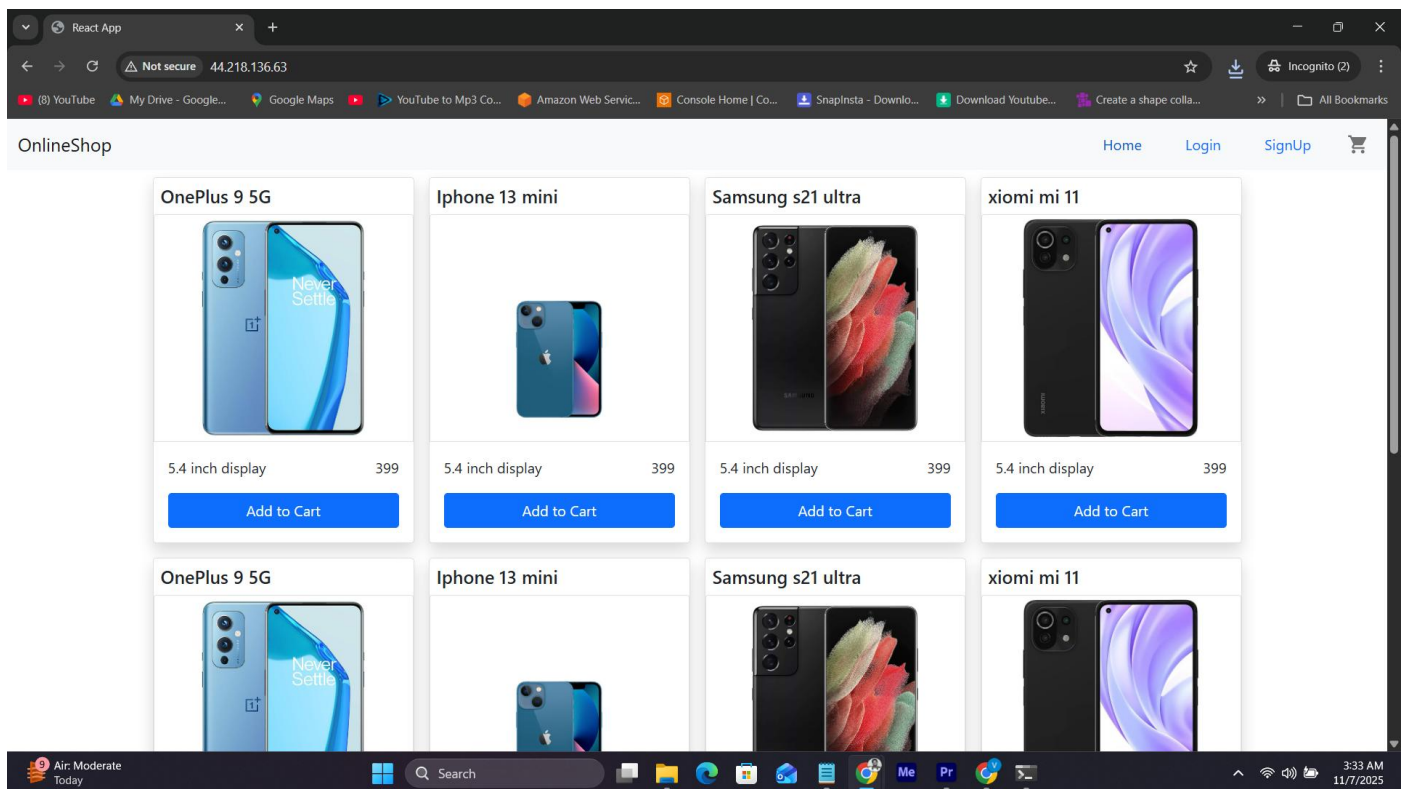
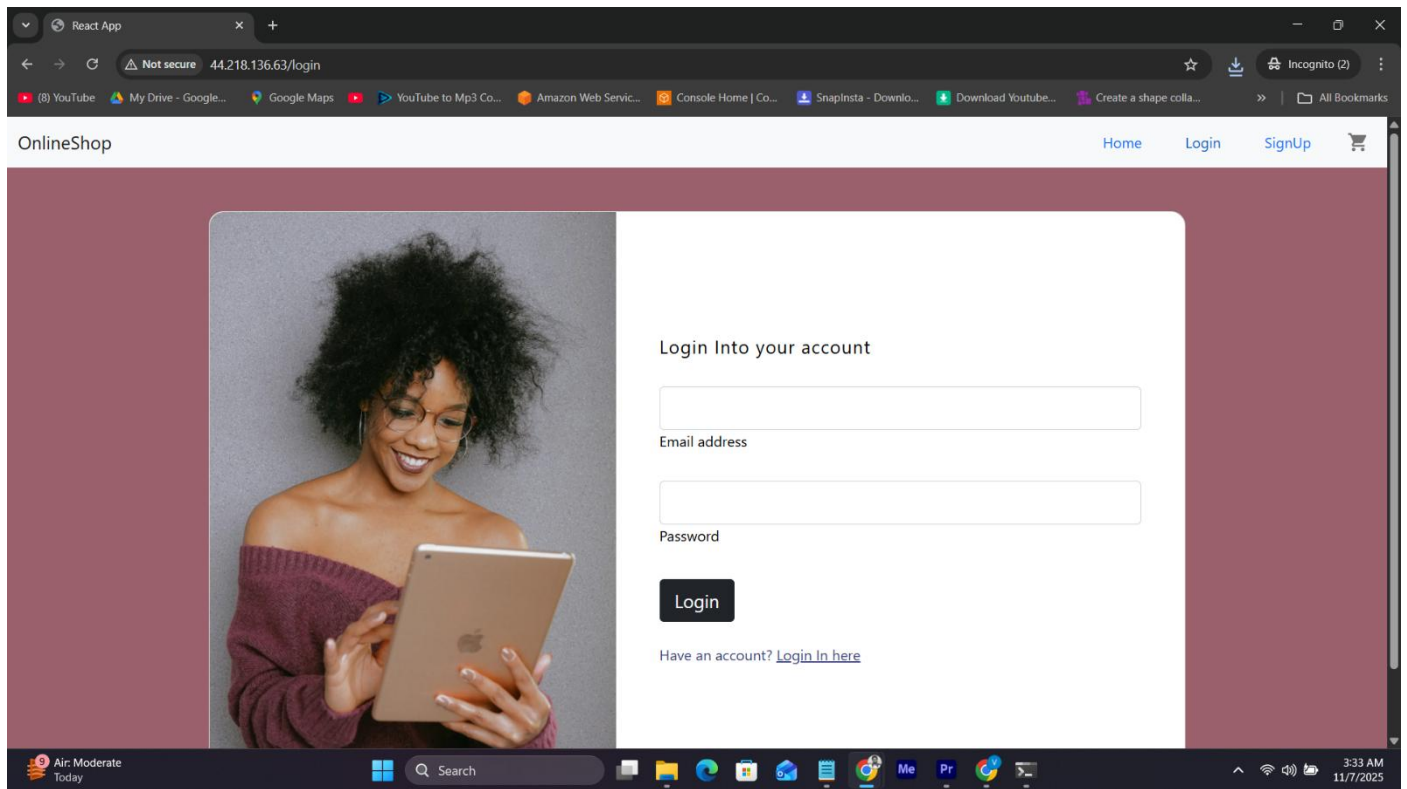
# Merge dev changes into main
git merge dev

# Push merged code to GitHub
git push origin main
Switched to branch 'main'
fatal: couldn't find remote ref main
Updating ef44730..9c281c8
Fast-forward
 .dockerignore      | 4 ++++
 .gitignore          | 4 ++++
 Dockerfile          |11 ++++++++
 Jenkinsfile         |64 ++++++++
 README.md           |1 +
 build.sh            |12 ++++++++
 deploy.sh           |22 ++++++++
 docker-compose.yml  |8 ++++++
 8 files changed, 126 insertions(+)
 create mode 100644 .dockerignore
 create mode 100644 .gitignore
 create mode 100644 Dockerfile
 create mode 100644 Jenkinsfile
 create mode 100644 README.md
 create mode 100755 build.sh
 create mode 100755 deploy.sh
 create mode 100644 docker-compose.yml
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0
remote:
remote: Create a pull request for 'main' on GitHub by visiting:
remote:   https://github.com/vigneshmarimuthu/React-App-CI-CD/pull/new/main
remote:
To github.com:vigneshmarimuthu/React-App-CI-CD.git
 * [new branch]    main -> main
ubuntu@ip-172-31-19-119: ~/devops-build$

```

The screenshot shows the Docker Hub interface for the repository `vigneshmarimuthu06/prod`. The page is divided into several sections:

- Header:** Docker Hub logo, navigation tabs (Explore, My Hub), search bar, and user profile (vigneshmarimuthu06).
- Repository Info:**
 - Repository name: `vigneshmarimuthu06/prod`
 - Last pushed: 1 minute ago
 - Repository size: 61.2 MB
 - Stars: 0
 - Downloads: 2
- Tags:**
 - Tags section: This repository contains 1 tag(s).
 - Table of tags:
- Repository overview:**
 - Repository overview section: An overview describes what your image does and how to run it.
- Build Cloud:**
 - Build with Docker Build Cloud section: Accelerate image build times with access to cloud-based builders and shared cache.
- Left Sidebar:**
 - Repositories
 - Hardened Images
 - Collaborations
 - Settings
 - Default privacy
 - Notifications
 - Billing
 - Usage
 - Pulls
 - Storage



✓ 11. Monitoring: Install Uptime Kuma (self-hosted)

- Uptime Kuma is a simple, open-source uptime monitor with alerts. It will only notify you when the app is down.

The screenshot shows the 'Add New Monitor' page in Uptime Kuma. On the left, there's a sidebar with a search bar and filters for Status, Active, and Tags. The main area is divided into sections: General, Notifications, Proxy, and HTTP Options. The General section has fields for Monitor Type (HTTP(s)), Friendly Name (Prod App), URL (https://44.218.136.63), Heartbeat Interval (60 seconds), and Retries (0). The Notifications section says 'Not available, please setup.' with a 'Setup Notification' button. The Proxy section also says 'Not available, please setup.' with a 'Setup Proxy' button. The HTTP Options section has a Method dropdown set to GET. A 'Save' button is at the bottom.

Uptime Kuma

Status Pages Dashboard

+ Add New Monitor

Add New Monitor

Select Search...

Status Active Tags

No Monitors, please [add one](#)

General

Monitor Type

HTTP(s)

Friendly Name

Prod App

URL

https://44.218.136.63

Heartbeat Interval (Check every 60 seconds)

60

Retries

0

Save

Notifications

Not available, please setup.

Setup Notification

Proxy

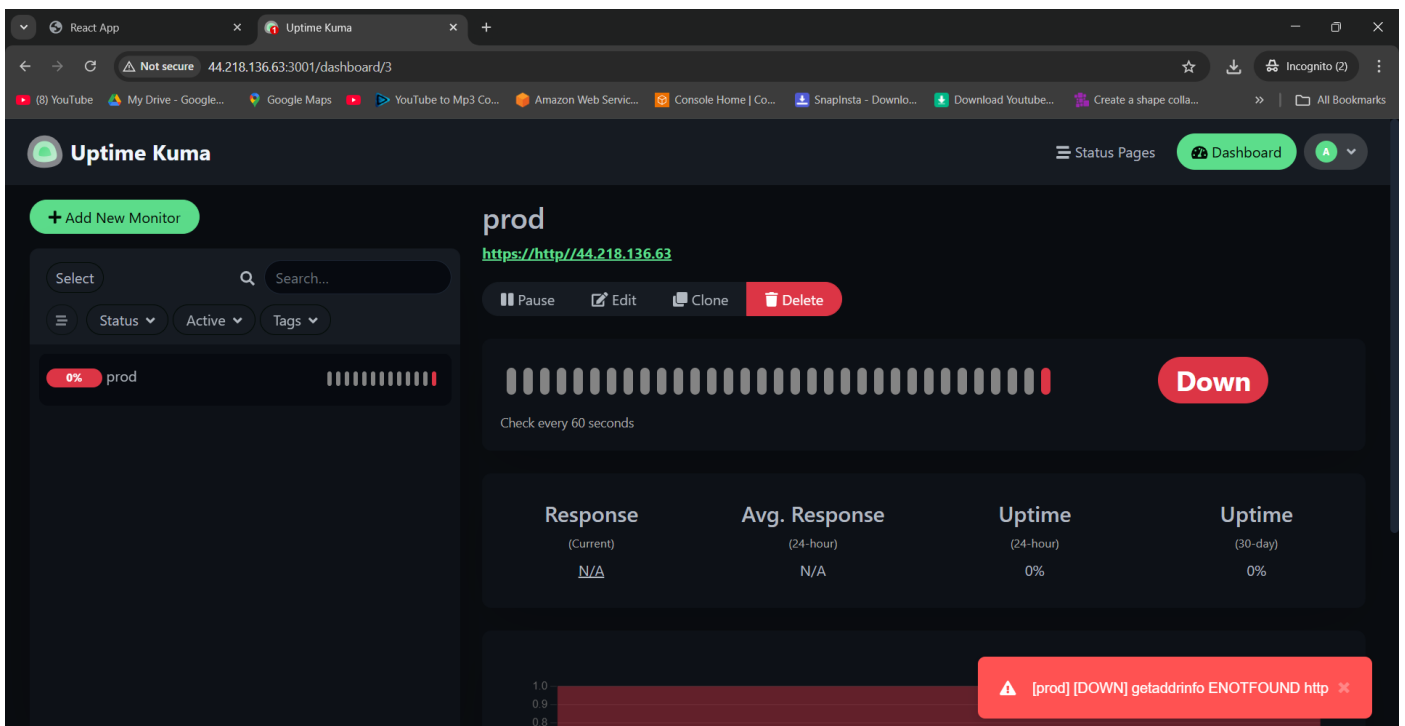
Not available, please setup.

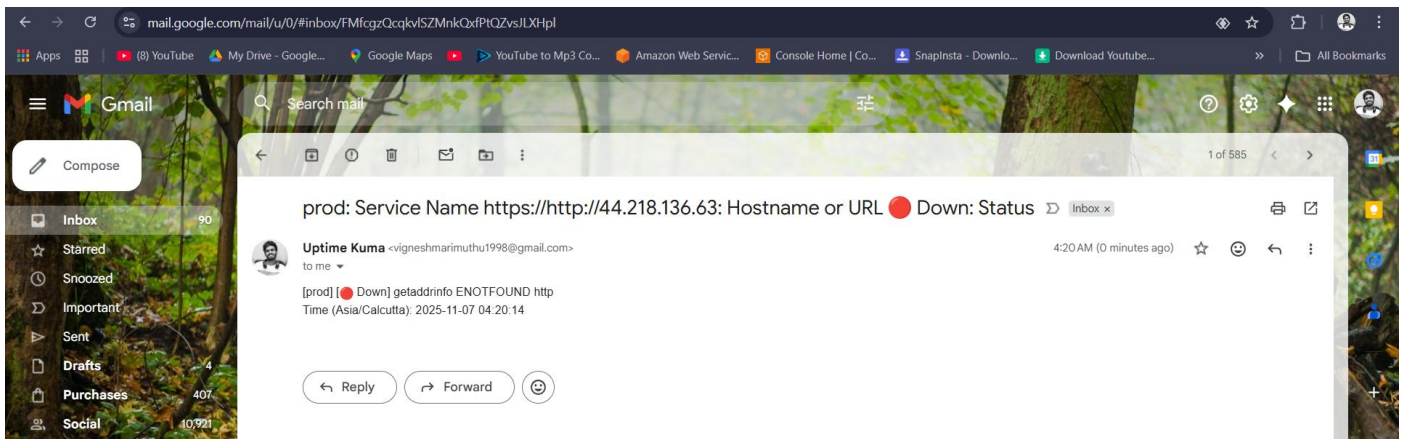
Setup Proxy

HTTP Options

Method

GET





- I checked Uptime Kuma and monitored it.
 - When I shut down Docker, it went down and I received a notification from the server. Everything was checked and it's all working.
-